

The number 10 shows that ₹ 25 must change hands for 10 times to arrange the transaction value of ₹ 250. This is clear that the calculation of T in real life is almost impossible and therefore, the above equation does not seem to be practical. Therefore, we have another version of the quantity theory i.e., *income version*. As per this version the transaction is converted to income equation by replacing T with income Y assuming that Y is a proxy of T. The logic behind this proxy is that T and Y are positively related though they are not equal, yet they are proportional. Thus, the quantity equation is

$$M \times V = P \times Y$$

Where,

M = Quantity of money i.e. money supply

V = Income velocity of money instead of transaction velocity.

P = Price level

Y = Real output

Relationship Between the Quantity Theory of Money and Demand for Money

From the transaction version of the theory, we know that

$$M \times V = P \times T$$

Think that there is a relationship between the volume of transactions T and the demand for money because the more value of T means more transactions and more transactions means the more demand for money. In this regard, Laidler¹ says that

“ though not stated as such by Fisher, this is equivalent to the following theory of the money market put in supply and demand terms. The demand for money depends upon the volume of the transactions to be conducted in the economy and is equal to a constant fraction of these transactions. Furthermore, the supply of money is given, and in equilibrium the demand for money must be equal to its supply” .

Thus, M can be replaced by M_d in equilibrium situation, therefore, the above equation should be

$$M_d \times V = P \times T$$

Moreover, if V and T are constants represented by $\bar{V}$ and $\bar{T}$, therefore,

$$M_d \times \bar{V} = P \times \bar{T}$$

Or

¹ Laidler, D. E. W. (1977). The Demand for Money: Theories and Evidence (2nd ed.). Dun-Donnelley Publishing Corp.

$$M_d = P \times \frac{1}{\bar{V}} \times \bar{T}$$

Let

$$\frac{1}{\bar{V}} = k$$

Therefore,

$$M_d = P \times k \times \bar{T}$$

$$M_d = k (P \times \bar{T})$$

Thus,

$$\text{Money demand} = k \times \text{volume of transaction}$$

Moreover, in classical frame work, we know that volume of transaction is equal to the Y, therefore,

$$\text{Money demand} = k \times \text{Income}$$

Criticism of the quantity theory

- a) British economist J.M. Keynes criticised the quantity theory of money in his book *The General Theory of Employment, Interest and Money*. He says that inflation is not a result of money supply but the aggregate demand. When at full employment level, the aggregate demand exceeds the aggregate supply, then there is inflationary gap. The increase in aggregate demand is independent of the money supply change.
- b) Nobel laureate Paul Krugman empirically found that the theory is not applicable during in a liquidity trap. Liquidity trap is a situation when the interest reaches to its minimum and further fall in the interest is not expected and monetary policy is ineffective.
- c) Austrian-American economist Ludwig von Mises criticised the theory by saying that the quantity theory of money does not focus on money demand side. That's a different matter that in equilibrium, the theory can help to determine the demand for money, but not in non-equilibrium situation.
- d) Some economists argue that the quantity theory of money is not theory in actual but the tautology.

5.4.2 The Cambridge Approach or Neoclassical Theory

Neoclassical theory is an improved version of the quantity theory of money. The credit of this theory goes to Alfred Marshall, A.C. Pigou, D.H. Robertson and J.M, Keynes. Since, all these economists were from the Cambridge University, therefore, the theory got popular as the Cambridge version of the quantity theory of money. These Cambridge economists argue that money is

mainly demanded for transaction purposes and people do not hold their incomes entirely in money form (i.e., cash at home) because money at home will not earn interest. Thus, only a portion (denoted by k) of the income is held at home for transaction purposes. In equation form,

$$M_d = k(PY)$$

where

Y = Real income

P = Price level

M_d = Money demand

Thus, according to Neoclassical theory of money, demand for money is directly determined, and money supply can be determined in equilibrium situation by keeping M_d equal to money supply. This is an opposite to the quantity theory of money in which money supply is replaced by money demand. Moreover, here k is known as Cambridge k , which is the reciprocal of the k in the quantity theory of money.

5.4.3 Liquidity Preference Theory or Keynes Theory of Demand for Money

The Liquidity Preference Theory, proposed by John Maynard Keynes, is a cornerstone of modern macroeconomics. It suggests that the demand for money is not just determined by transactions but also by the desire of individuals to hold liquid assets for unforeseen contingencies. Keynes accepts that the main motive of money is its use as medium of exchange, but he also recognises that there are two other motives too. According to Keynes, individuals have a preference for liquidity, meaning they value money not only for its ability to facilitate transactions but also for its security and ease of conversion into other assets. According to Keynes, money demand $M(d)$ is the summation of the following three components:

- a) Transaction motive M_d^T
- b) Precaution motive M_d^p , and
- c) Speculative motive M_d^{sp} .

The detail of each component is as follows:

a) Transaction motive

People do various small and big transactions in their day-to-day life. For example: they purchase clothes, they buy cars, they buy cereals, they buy flour, they pay interest, they repay their loans, they pay salaries, they pay electricity bills, etc. All these transactions require money and this demand is met by the income. Keynes confined the transaction motive as

the necessity of cash to fill the gap between receipts and planned regular payments (e.g., school fees, electricity bills). Higher the level of income means the higher the level of transactions for goods and services requiring more money. Therefore, the transaction demand is directly related to the income. In equation form:

$$\text{Nominal money demand} = M_d^T = \alpha(PY)$$

Or

$$\text{Real money demand} = \frac{M_d^T}{P} = \alpha Y$$

where, α shows how much of the nominal income (PY) is demanded for transaction purposes. The above equation shows that there is positive relationship between the money demanded for transaction purpose and the nominal income. Keynesians argue that money demand for transaction purpose is interest-inelastic i.e., interest rate does not affect the money demand, however, some economists argue that beyond a certain level of interest rate (i_3 in the diagram), the money demand gets interest-elastic because such high interest rate is perceived to be high opportunity cost of holding cash balance.

b) Precautionary motive

Future is uncertain. Nobody knows what is going to happen in future, therefore, to meet such unforeseen happenings people demand money. Precautionary motive is one of the reasons people save their income to be used in odd circumstances like death of a family member. The demand for precautionary purpose is also directly related to income. In equation form:

$$\text{Nominal money demand} = M_d^p = \beta(PY)$$

Or

$$\text{Real money demand} = \frac{M_d^p}{P} = \beta Y$$

Here, β is a constant that shows how much portion of the nominal income (PY) is demand for precautionary purpose. Keynesians think that precautionary purpose demand is also interest-inelastic but some other economists think that beyond a certain level of interest rate it will be interest-elastic. Thus, the behaviour of the precautionary purpose demand curve is the same as that of the transaction purpose demand.

c) Speculative Motive

Although, speculation can be found in any market whether financial market or otherwise, Keynes' explanation for speculation demand for money revolves around bond market. Speculation means making profits

due to price fluctuation of an asset (e.g., a share, or bond). For example, if you purchase a bond for ₹ 1000 today and the price reaches to ₹ 1200 after a year, then you will earn ₹ 200 as profit due to price change. Apart, ₹ 200, you will earn some interest also (generally known as coupon payment) for holding the bond for 1 year. The profits due to price fluctuations are known as capital gains. With regard to a perpetual bond (i.e., a bond having infinity maturity period, the value of a bond is calculated using the following formula:

$$V = \frac{C}{i}$$

where,

V = Value of a bond.

C = Coupon i.e., interest on the bond.

i = Market interest rate or discounting rate.

The formula states that there is an inverse-relationship between the value of a bond and the interest rate. As the interest rate rises, the value of the bond falls down and as the interest rate falls, the value rises up. Keynes argues that, at any time, there is a level of interest rate called normal interest rate and as the interest rate goes above it, then an individual expects the interest rate to fall and when the interest rate is below it, then the individual expects the rise in interest rate. When an individual expects the interest rate to fall (known as a bearish expectation), then he will earn capital gain and interest due to holding period of the bond. When he expects the interest rate to rise (known as a bullish expectation), then he will suffer capital loss, but still he will earn coupon (because the coupon is fixed by the issuer). As long as the coupon exceeds the capital loss, then he will hold the bond and all the wealth will be converted into bonds. If the capital loss exceeds the coupon, then he will not hold bonds, but he would prefer his total wealth into money form i.e., cash. At aggregate level, the weightage of each individual's holdings (in money and bond) is not significant and the aggregative speculative demand for money is a smooth function of the interest rate. According to Laidler, the speculative demand for money function can be shown by the following equation where W is real wealth.

$$M_d^{sp} = (P \times W) \times f(i)$$

Therefore, total money demand can be expressed

$$M_d = \alpha Y + \beta Y + (P \times W) \times f(i)$$

Alternatively,

$$M_d = (\alpha + \beta)Y + (P \times W) \times f(i)$$

Thus, money demand is a function of Y , W , and interest rate.

Some textbooks (e.g., Principles of Macroeconomics by N.G. Mankiw) summarize the liquidity preference theory by the following equation:

$$\text{Demand for nominal money} = M_d = PL(i, Y)$$

Alternatively,

$$\text{Demand for real money} = \frac{M_d}{P} = L(i, Y)$$

Criticism of liquidity preference theory

- a) Keynes' segmentation of the demand for money into three categories is impractical as it assumes individuals maintain distinct purses for different purposes, whereas in reality, people typically manage their finances from a single purse that serves all their needs.
- b) Some economists have questioned the empirical validity of liquidity preference theory, arguing that it fails to explain observed fluctuations in interest rates and money demand accurately. Empirical studies have shown that factors beyond liquidity preference, such as inflation expectations, economic growth, and central bank policies, also influence interest rates and money demand.
- c) The theory assumes that all individuals have the same expectations about future interest rates and income. However, in reality, people have diverse expectations, leading to variations in their demand for money. This heterogeneity is not adequately accounted for in liquidity preference theory.

5.4.4 Baumol – Tobin Theory of Transaction Demand for Money

This theory was developed by William Jack Baumol (1952) and James Tobin (1956) independently, however, their results were found to be similar, that is why, this theory is commonly termed as Baumol – Tobin theory for transaction demand for money. The theory is also known as the Baumol-Tobin Model of Cash Management. The theory is similar to the concept of economic order quantity (EOQ) in inventory management. Just as the economic order quantity is the quantity of the material which minimizes the total cost consisting of purchase cost, order cost, and carrying cost, similarly, the Baumol-Tobin theory helps us to find the level of cash (demand for money for transaction purpose) which an individual should hold to minimize total cost consisting of transaction cost and operational cost. How? Although, the theory can be easily understood by minimizing the cost of demand for money with reference to investment in bonds, yet to keep the theory simple, we assume that an individual is willing to spend ₹ Y annually by withdrawing from his bank account carrying interest i .

If ₹ Y are withdrawn in one shot at the beginning of the year and spent gradually at constant rate, then at the end of the year the balance should be zero, therefore, his average holding should be $Y/2 = [(opening\ balance + closing\ balance)/2]$.

If he makes half withdrawal at beginning of the year, the opening balance should be $Y/2$ and it will be spent over the first 6 months and another half balance will be withdrawn and spent over the next 6 months, therefore, the average holding should be $Y/4$. On the same pattern, if he makes Q number of trips to the bank, then at the beginning he withdraws Q/N money, then the average holding should be $Q/2N$.

Now, to withdraw the cash from bank, he will have to incur two types of costs given below:

- a) Cost of transaction – It includes the cost incurred to conclude the transaction of withdrawing cash e.g., travel cost incurred to visit the bank.
- b) Opportunity cost – This is nothing but the interest lost, when the cash is withdrawn from the bank because lower balance will earn lower interest.

If there are Q number of visits to bank and cost of per visit (or transaction cost) is b , then the cost of visit should be :

$$bQ$$

Moreover, if i is the interest rate, then the opportunity cost should be

$$\frac{iY}{2Q}$$

Therefore, total cost (TC) should be

$$TC = \frac{iY}{2Q} + bQ$$

Now, to minimise the total cost, we use differential calculus and find out the first-order condition of minimization i.e.,

$$\frac{d\phi}{dQ} = 0$$

$$-\frac{iY}{2Q^2} + b = 0$$

$$b = \frac{iY}{2Q^2}$$

$$Q^2 = \frac{iY}{2b}$$

$$Q = \sqrt{\frac{iY}{2b}}$$

Further, the average money holding is

$$A = \frac{Y}{2Q} = \frac{1}{2} \times \frac{Y}{\sqrt{\frac{iY}{2b}}}$$

$$A = \frac{1}{2} \times Y \sqrt{\frac{2b}{iY}}$$

$$A = \sqrt{\frac{bY}{2i}}$$

The above formula gives us the average money holding as per the Baumol-Tobin theory. The following illustrates the formula: Mr. M wants to withdraw from his bank balance of ₹ 1,00,000. The interest rate is 5% per annum and transaction cost (or cost of per visit) ₹ 4, then the optimum number of visits and average money holding can be calculated as follows:

$$Q = \sqrt{\frac{0.05 \times 1,00,000}{2 \times 4}} = 25$$

$$A = \sqrt{\frac{4 \times 1,00,000}{2 \times 0.05}} = \sqrt{40,00,000} = 2,000$$

The above results can be verified through the following table and Figure 2. This is obvious that total cost curve TC is minimum at Q=25.

Number of visits (Q)	Average monthly holding = $Y/2Q$	Visit cost = $Q \times ₹4$	Opportunity cost = Average monthly holding $\times 5\%$	Total Cost
5	10000	20	500.00	520.00
7	7143	28	357.14	385.14
10	5000	40	250.00	290.00
13	3846	52	192.31	244.31
16	3125	64	156.25	220.25
19	2632	76	131.58	207.58

22	2273	88	113.64	201.64
25	2000	100	100.00	200.00
28	1786	112	89.29	201.29
31	1613	124	80.65	204.65
34	1471	136	73.53	209.53
37	1351	148	67.57	215.57
40	1250	160	62.50	222.50
43	1163	172	58.14	230.14
46	1087	184	54.35	238.35
49	1020	196	51.02	247.02

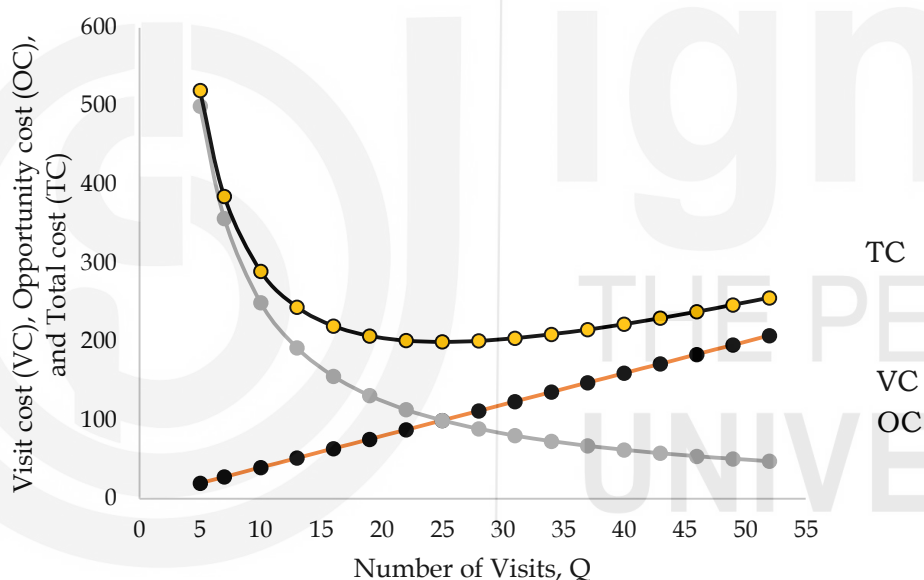


Fig. 5.2 : Optimum Number of Visits

Criticisms to the Baumol – Tobin Theory

- The theory assumes that the money is spent at constant rate, but in real life this assumption is unrealistic because people do not know with certainty how much they need to spend.
- It is not always easy to quantify the cost of transactions.

5.4.5 Tobin's Theory of Speculative Demand for Money

James Tobin published a paper², in which he says that individuals make the portfolio of money as well as bond, and the speculative demand for money is

² Tobin, J. (1958). Liquidity Preference as Behavior Towards Risk. The Review of Economic Studies, 25(2), 65. <https://doi.org/10.2307/2296205>

a matter of uncertainty not the expectations about the interest rates as opposed to Keynes who used to think that individuals hold either all cash or all bonds depending upon their expectations of interest rate. Let us understand the view of Tobin. Suppose that V represents the value of the bonds and r is the rate of return, then total return should be

$$R = rV$$

Moreover, if θ is rate of risk, then the total risk should be

$$\delta = \theta V$$

Combining these two equations we get

$$R = \frac{r}{\theta} \delta$$

The ratio $\frac{r}{\theta}$ should be treated as a constant β , then we have

$$R = \beta \delta$$

The last equation shows a linear relationship between R and δ as shown in Figure 5 through the line OZ . The line shows that higher the level of risk (shown on the horizontal axis in panel (a)) leads higher return (shown on the vertical axis in panel (a)).

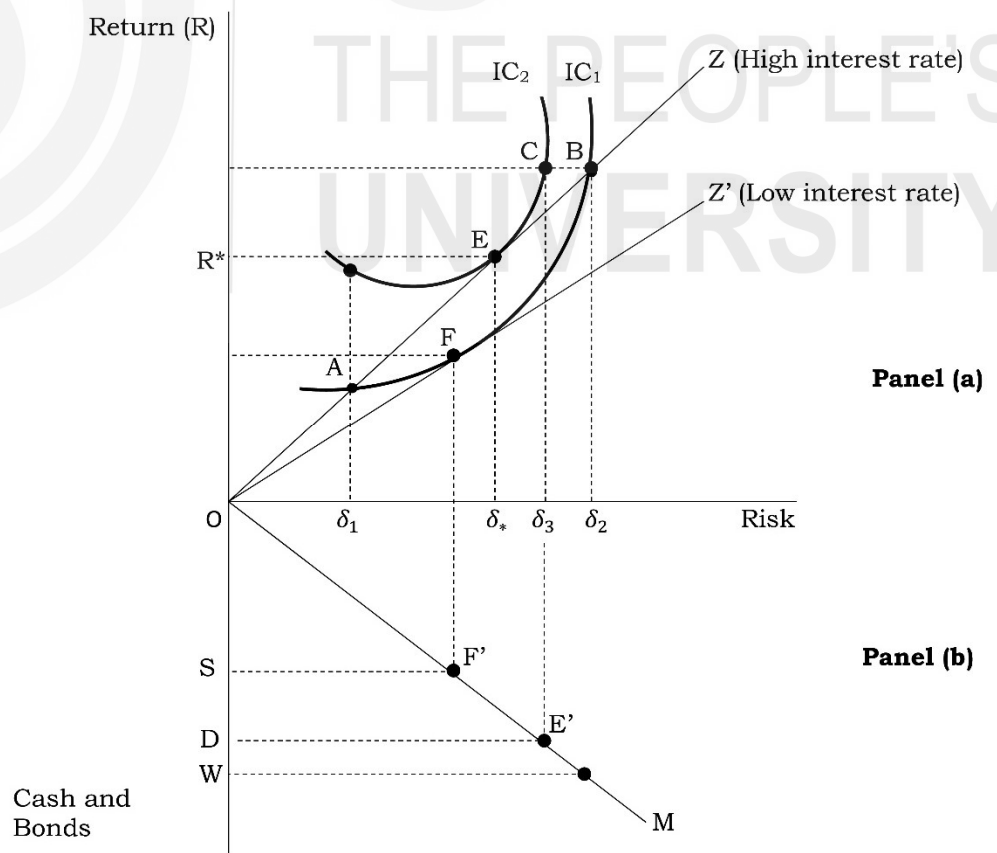


Fig. 5.3: Tobin's Portfolio Optimization and Bonds and Cash Weights Determination

In panel (a), IC_1 and IC_2 are two indifference curves showing different levels of utility for various combinations of risk and return. From the properties of the indifference curve, we know that higher the level of indifference curve is preferable, therefore, an investor would prefer IC_2 in comparison to IC_1 . To prove this, let us consider that the investor chooses point A on IC_1 where the risk is δ_1 , but with the same amount of risk, a higher return on IC_2 is available, but that return is not achievable because it does not fall on OZ. Similarly, at point B, the risk is δ_2 , but the same return at point C on IC_2 is available with lower risk δ_3 . However, point C cannot be achieved because it also does not fall on OZ. Thus, any point below IC_2 will give the investor lower utility because of the opportunity to achieve the higher indifference curve. Thus, the investor will maximise his utility when the OZ line touches the indifference curve at point E. To achieve this point E, the investor would need investment in bonds equal to OD out of total wealth W as shown on the vertical axis in panel (b). Thus, remaining amount WD is cash remaining with the investor. The line OM shows such various combinations of bond and cash available with the investor for a given point in panel (a). Now, if the interest rate falls, then the OZ line will rotate clockwise, say, OZ' touching IC_1 at point F. Thus, new equilibrium (point of optimization) point is F at which investment in bonds is OS while the cash is WS. This is obvious that $OS < OD$ implying that fall in interest rate causes the investment in bonds to fall. Therefore, according to Tobin, people combine bonds and cash in a ratio to make a portfolio depending upon their point of optimization.

5.4.6 Friedman's Quantity Theory of Money

Milton Friedman, an American economist cum statistician and the 1976 Nobel prize winner, developed his theory of demand for money. Infact, Friedman regarded his theory as a restatement of the quantity theory of money, that is why, his theory sometimes called modern quantity theory of money. According to Friedman, money demand can be studied in the same manner as the demand for any asset. Thus, the Friedman's 1970 version of the demand for money function is as follows:

$$M_d = f(Y_p, W, i_m, i_b, i_e, \pi_e, u)P$$

where,

P is the price level. The logic of inclusion of the price level is that changes the purchasing power of the money. If P increases, then it reduces the real money balance i.e., the purchasing power and to compensate it people demand more money.

Y_p is permanent income which is used as a proxy of human wealth (i.e., labour earnings) because wealth cannot be directly measured. Permanent income is weighted average of the past and present incomes. Higher the level of permanent income means higher the money demand.

W is non-human wealth e.g., land or machines.

i_m is the return on money itself and if it increases, then the people demand more money.

i_b is the return on bond. Its decrease or increase result in capital gain or loss, therefore, it is negatively related to the money demand.

i_e is the return on equities. It is negatively related to money demand because higher the return on equities will attract the investors to earn dividends and, therefore, less money demand.

π_e is the expected inflation rate. When people expect inflation in future, then their purchasing power will decline, therefore, they would like spend or invest money before their expectations come true. Therefore, there is a negative relationship between the money demand and expected inflation rate. Expected inflation can be calculated as follows:

$$\pi_e = \frac{1}{P} \frac{dP^e}{dt}$$

u is a variable representing all those factors not considered in the function e.g., taste and preferences. Its relationship with the money demand may be in either of the directions.

Check Your Progress 2

- 1) Although the quantity theory of money is not a theory of demand, then how can it be used as a theory of demand for money?
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- 2) What are the three motives for which money is demanded ?
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- 3) What is Tobin's theory for demand for money?
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- 4) What is the usefulness of the Baumol – Tobin model?
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